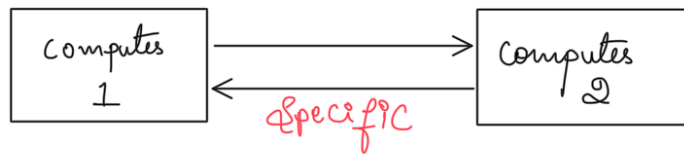
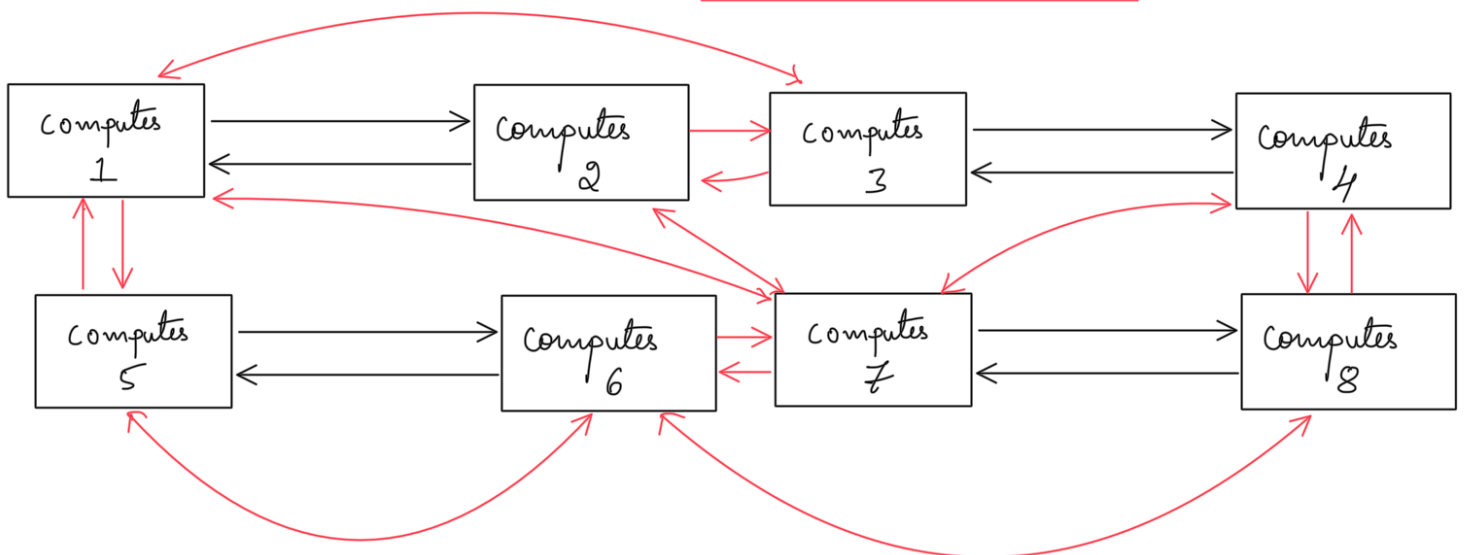


In 1960 Between Two Machines for communication they have to design custom Wires, protocols that communication is very specific with two machines



Because of this specific set of rules communication with other computers is not possible.

So the communication needs to be moved to more generic approach which makes communication easier.



using generic set of rules communication becomes more easier. Every computer use same type of protocols wires and instructions

So they can share information very easily

ARPA NET & TCP/IP History

- In the 1960's the U.S Department of Defense's ARPA created ARPANET. the first computer network
- It introduced packet switching - Instead of reserving a full line like telephone calls (circuit switching)

- data was sent in small "packets"
- ARPANET also had a decentralized design (no single failure point) important during the cold war

Fun fact : The very first message sent in 1969 was "LO" (meant to be Login) before the system crashed

- In the 1980's, the TCP/IP protocol suite became the global standard, connecting different networks into one internet
- Vint Cerf and Bob Kahn are called the "fathers of the Internet" for their work on TCP/IP
- In the 1990s Tim Berners-Lee invented the World Wide Web (HTTP + HTML) → first websites

MAC Addressing & NIC

- Every device has a Network Interface Card (NIC) → Ethernet adapters or Wi-Fi chip.
- Each NIC has a Mac address (unique 48 bit hardware ID, Eg 00:1A:0B:3C:4D:5E)
- Used for local network communication (LAN)

How it works

- your laptop → sends data → frames it with the destination MAC
- Switch → forwards it only to the correct device (like a postman reading house numbers)
- ARP (Address Resolution protocol) finds which MAC belongs to which IP (like a phone book)

MAC address helps to identify the computer

How it works

- Your laptop → sends data → frames it with the destination MAC
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to which IP (like a phone book)



Packet Switching & Data Encoding

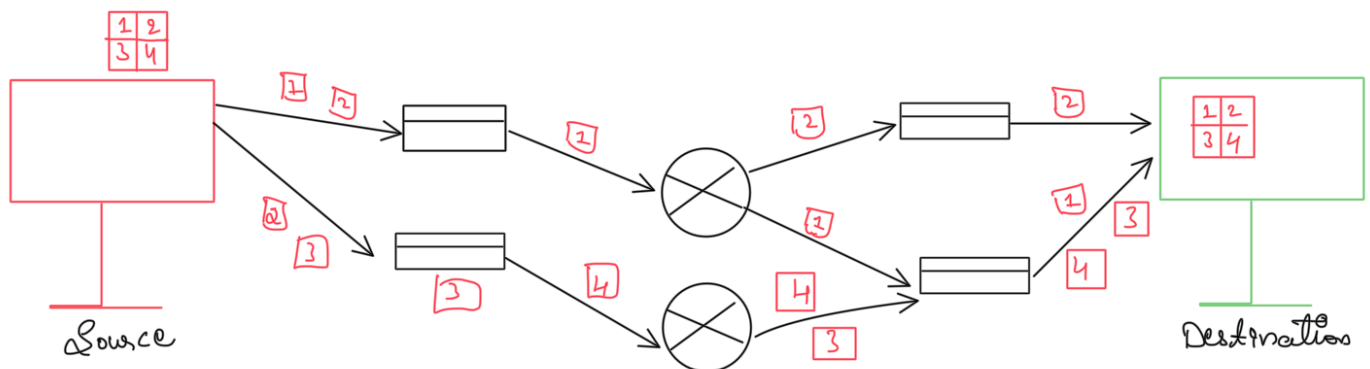
Circuit Switching (old Telephony)

- A dedicated line for the entire call
- Example landline phones

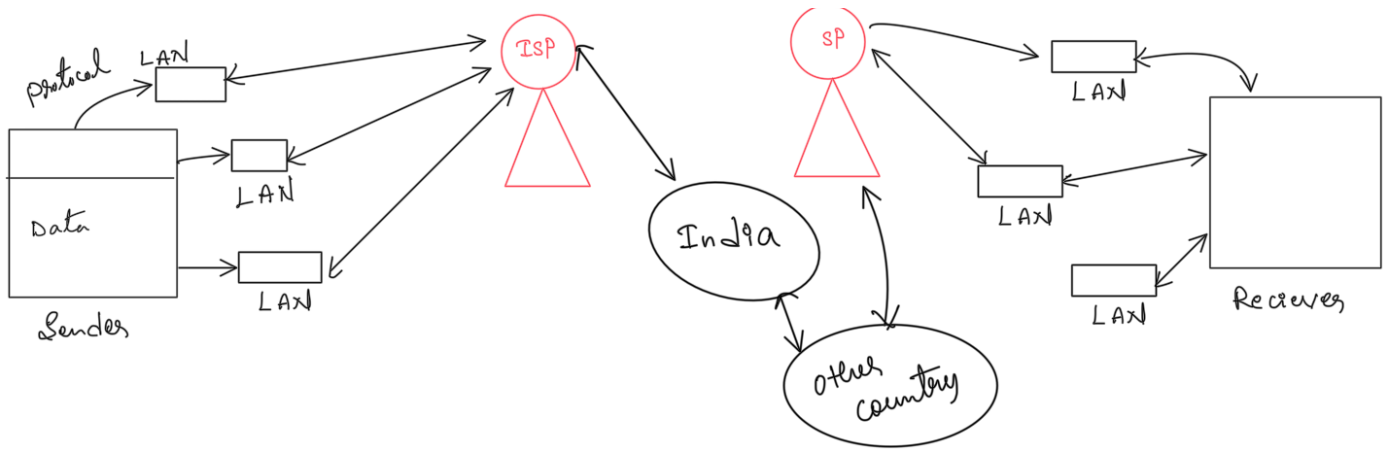
Packet Switching (Internet)

- Breaks data into small packets
- Each packet has Source, destination, sequence number
- Packets may take different routes, then get reassembled at the destination.

Analogy Think of Sending a book by courier - instead of shipping the whole book at once, you send it Page By Page in different trucks. At the destination, Pages are reassembled.



Data get divided into small packets and sent to destination through routers & switches



HTTP → At Application level
GET Request

In High level

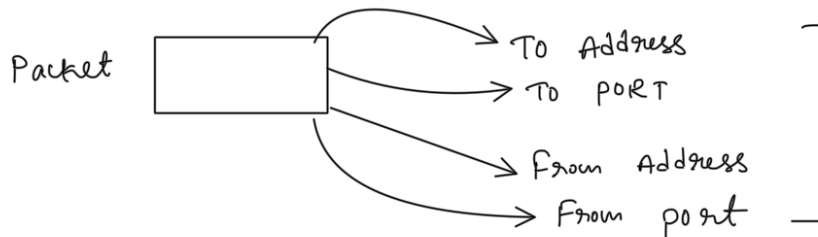
URL, Header

1
packets

2
packets

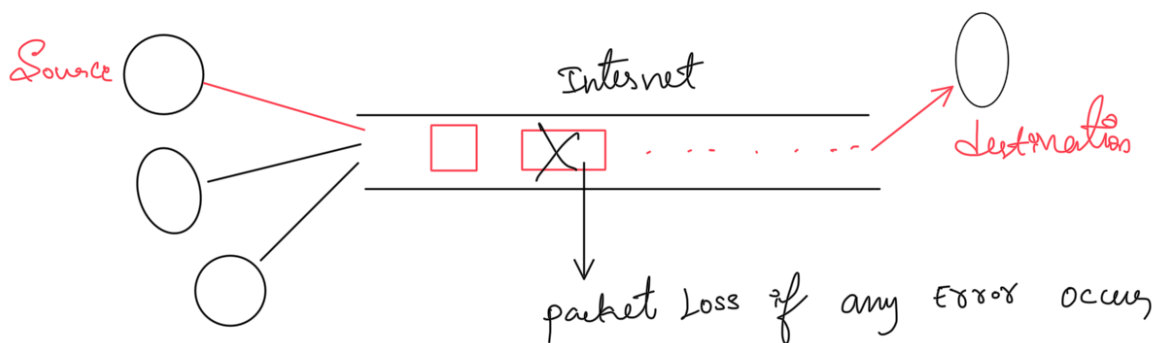
3
packets

Sent over Internet
going as 1, 0
1 0 1 0 0 1 0 1



Packet Have all this
Information.

The Information in the packet is Sent as electrical signals
Ex "10 100 10" if any one error occurs that packet will
be corrupted that's called "packet Loss"



Tomorrow I will continue my learning on Networks More